# Section 1.3 — Functions and Relations — Answer Key

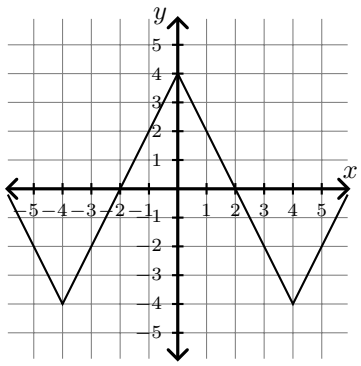
Math 103: College Algebra

## 1 Understanding Relations

Let $R = \{(2,5),(3,7),(-8,-2),(9,1)\}$. Decide whether each statement below is true.

1. $2R5$ – true
2. $-7R8$ – false
3. $7R3$ – false
4. $-8R-2$ – true

Let $S$ be the relation graphed below. Decide whether each statement below is true.




5. $0S4$ – true
6. $-4S4$ – false
7. $3S-2$ – true
8. $4S-4$ – true

Let $T = \left\{(x,y) | x^2+y^2=1\right\}$. Decide whether each statement below is true.

9. $0T1$ – true
10. $-1T0$ – true
11. $-1T-1$ – false
12. $0.6T-0.8$ – true

## 2 Domain and Range of a Relation

For each relation below, describe the domain and range.

13. $\operatorname{Dom} Q = \left\{-1, 1, \frac{11}{8}, 2\right\}$
    $\operatorname{Rng} Q = \left\{\frac{5}{3}, 2, 3, 35\right\}$
14. $\operatorname{Dom} R = \left\{1\right\}$
    $\operatorname{Rng} R = \left\{2,3,4,5\right\}$
15. $\operatorname{Dom} S = \left\{2,3\right\}$
    $\operatorname{Rng} S = \left\{-1,7,9\right\}$
16. $\operatorname{Dom} T = \left\{-2, 2.2, \pi, 11\right\}$
    $\operatorname{Rng} T = \left\{1\right\}$
17. $\operatorname{Dom} Q = [-4,2]$
    $\operatorname{Rng} Q = [-4,2]$
18. $\operatorname{Dom} V = [-2,1]$
    $\operatorname{Rng} V = [-2,2]$
19. $\operatorname{Dom} W = [-3,-2] \cup [-1,4]$
    $\operatorname{Rng} W = [-3,2]$
20. $\operatorname{Dom} Z = [-3,3]$
    $\operatorname{Rng} Z = [-3,1] \cup 2$

## 3 Identifying Functions

State whether each relation is a function. If not, give two pairs that show why not.

21. Function
22. Not a function: $(2,3)$ and $(2,4)$ have $x=2$.
23. Function
24. Not a function: $(-1,-1)$ and $(-1,1)$ have $x=-1$.
25. Not a function: $(0,4)$ and $(0,-4)$ have $x=0$. (Other pairs are possible.)
26. Function
27. Function
28. Not a function: $(2,1)$ and $(2,-1)$ have $x=2$. (Other pairs are possible.)

For each relation below, give two pairs $(x, y)$ that show that the relation is *not* a function. As a hint, remember that $x^2$ and $(-x)^2$ always give the same (positive) result.

29. $(2, 2)$ and $(2, -2)$ have $x = 2$. (Other pairs are possible.)

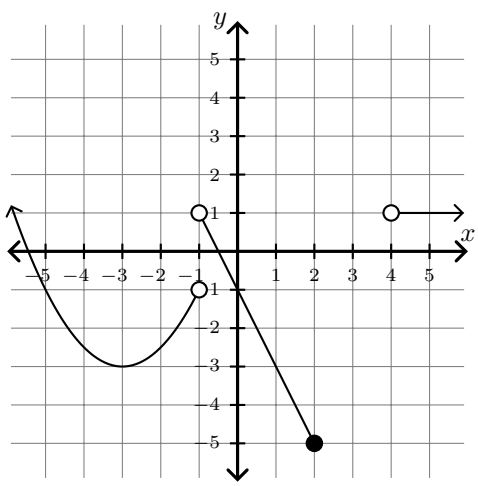
30. $(1, -1)$ and $(1, 1)$ have $x = 1$. (Other pairs are possible.)

## 4 Function Notation

Let $f = \{(-2, 5), (3, -2), (4, 7), (8, 9)\}$. Find each of the following.

31. $f(4) = 7$
32. $f(-2) = 5$
33. $f(9)$ is undefined
34. $f(3) = -2$

The function $g$ is graphed below. Find each of the following.




35. $g(-3) = -3$
36. $g(-1)$ is undefined
37. $g(1) = -3$
38. $g(3)$ is undefined

## 5 Equations of Functions

Write each function in relation notation.

39. $f = \{(x, y) \mid y = 2x - 5\}$
40. $g = \{(x, y) \mid y = x^2 - 7x + 12\}$
41. $h = \left\{(x, y) \mid y = \dfrac{3x - 5}{2x + 1}\right\}$

Each relation below is a function. Rewrite in function notation. You may need to first solve for $y$.

42. $R(x) = x^3 - x$
43. $S(x) = \frac{2}{5}x - 2$
44. $T(x) = x^2 - 3$

Let $k(x) = \dfrac{4x^2 + 2}{x - 7}$. Find each of the following. Give your answer as a fraction.

45. $k(2) = -\frac{18}{5}$
46. $k(-3) = -\frac{18}{5}$
47. $k(7)$ is undefined
48. $k(0) = \frac{2}{7}$

49. $\operatorname{Dom} k = \{x \mid x \neq 7\}$